LETTERS TO NATURE

An Alaskan Jurassic Palaeomagnetic Pole and the Alaskan Orocline

As part of an effort to determine whether or not the curvature of the Alaskan orocline, as outlined by the Alaska Range and certain other geological features such as the Denali fault system, is of primary or secondary origin, oriented samples of Jurassic age were collected from sixteen sites in four areas along the arcuate trend of the oroclinal bending[1,2] (Fig. 1). The geographic spread of the sites is about 1,100 km. The rock types sampled include fine-grained sandstones and siltstones and granitic intrusives which have variable magnetic intensities and stabilities. Age control for each of the sedimentary sites was derived from fossil evidence[3–5], and the ages of the batholiths were determined radiometrically[6]. All the samples were measured using a slow spinner magnetometer, and were demagnetized in a two-axis tumbler[7] in a 60 Hz alternating field. The demagnetization level used was selected on the basis of visual inspection of the demagnetization curves and the apparent stability of the data against further demagnetization.

Data were rejected from those sites for which the precision parameter k (ref. 8) was less than 5, and two other sites were also ignored, one because the resulting virtual geomagnetic pole was more than 90° away from the overall mean, and the other because it was considered still to have a secondary component after demagnetization. Nine sites were considered stable and representative of the Jurassic palaeofield (Table 1).

A test of the orocline hypothesis requires a comparison of the palaeomagnetic directions along the length of the arcuate trend both amongst themselves and with respect to a fixed point outside the system relative to which the bending must have taken place. The scatter of the data, together with the fact that the sites are all essentially on one limb of the orocline, made it impossible to test the oroclinal model by internal consistency of the palaeomagnetic declinations. It was originally hoped that the data could be compared with the Jurassic pole for North America as an external reference point. As can be seen from Fig. 2, however, the discrepancy between the southern Alaskan Jurassic pole and that obtained for North America east of the Sierra Nevada Mountains[9–11] cannot be explained by the bending of the Alaskan orocline because it is in the wrong direction.

If the Alaskan Jurassic data are interpreted in terms of the common assumption that the geomagnetic field is basically dipolar when averaged over several million years, then a palaeogeographic reconstruction based on the Alaskan pole would place Alaska about 18° south of its present position, and rotated counterclockwise by 52°. Accordingly, the continental margin of the Alaska Peninsula would have been approximately parallel to the Oregon–Washington continental margin. The palaeolongitude of southern Alaska cannot, however, be determined from a single palaeomagnetic pole.

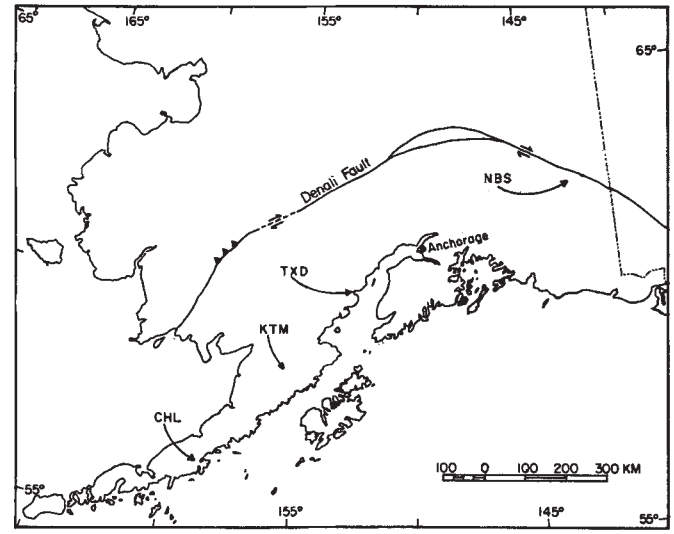


Fig. 1 Map showing sampling localities in southern Alaska. CHL, Chignik Lake; TXD, Tuxedni Bay; KTM, Katmai Area; NBS, Nabesna Area.

"Unfolding" the orocline shifts the mean pole south and west. This does not affect the palaeolatitude of Alaska, but requires a much greater clockwise rotation (about 100°) to have taken place between the Jurassic and the present.

It was not possible to compare the southern Alaskan data with pole positions derived from sites to the west of the Sierra Nevada Mountains, because of the lack of internal consistency of the west coast data[12–15]. The spread of palaeomagnetic directions derived from sites on the west coast suggests a complicated tectonic history for the whole western margin of the North American plate, thus precluding any long range comparison of data.

Our data point to a much more southerly location for at least parts of Alaska in Jurassic times, and this must be accounted for in any tectonic history of the area. The concept

Table 1 Palaeomagnetic Directions and Pole Positions

Site	Location	N	Direction				Pole position			
			D	I	$A95$	K	Lat	Elong	$A95$	K
CHL-1	56.3° N, 201.2° E	10	1	70	6.8	51.2	89	339	10.8	21.0
CHL-2	56.3° N, 201.2° E	10	62	61	13.5	13.8	49	282	19.5	7.1
CHL-4	56.3° N, 201.2° E	9	90	32	23.1	5.9	15	280	19.3	8.1
CHL-5	56.3° N, 201.2° E	12	81	39	16.6	7.8	23	284	18.0	6.8
TXD-1	60.1° N, 207.4° E	15	14	53	15.2	7.3	65	0	15.2	7.3
TXD-3	60.3° N, 207.1° E	10	36	45	18.4	7.8	50	336	20.5	6.5
TXD-4	60.2° N, 207.4° E	9	71	66	15.8	11.5	52	276	20.4	7.3
NBS-1	62.6° N, 216.9° E	14	63	65	7.4	30.2	54	297	10.6	15.1
NBS-3	62.6° N, 216.9° E	11	26	40	4.7	93.8	47	1	4.8	91.4
Mean		9					54	303	20.3	7.4

N, number of samples per site; D and I, declination and inclination; $A95$, 95% confidence circle about mean; K, Fisher parameter; Lat and Elong, north latitude and east longitude. Rock types: CHL-1, 2, 4, 5, fine-grained sandstone; TXD-1, fine-grained sandstone; TXD-3, quartz diorite; TXD-4, siltstone; NBS-1, 3, fine-grained sandstone. All results after demagnetization in an alternating field with a peak of 95 oersted (except TXD-3, 190 oersted).

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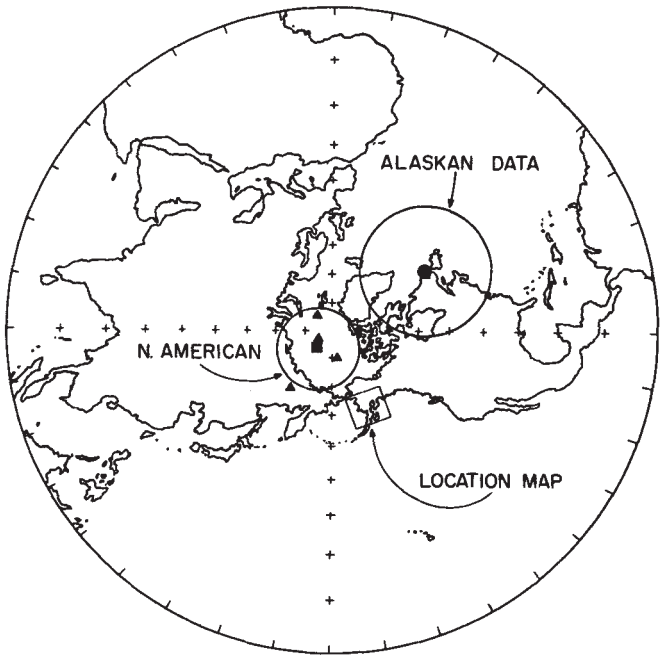


Fig. 2 Jurassic pole positions. ●, Mean of Alaskan data; ■, mean of North America Jurassic poles east of the Sierra Nevada Mountains; ▲, individual poles used to determine the Jurassic pole for North America. Circles are 95% confidence limits.

of Carey's Alaskan orocline would seem to be untenable on the basis of the additional rotation required.

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Rare Earth Element Distributions in Equatorial Mid-Atlantic Ridge Gabbros

THE crustal sections exposed in the fracture zones of the equatorial Mid-Atlantic Ridge afford the deepest direct sampling of the crust. The gabbros, amphibolites and greenschists underlie basalts and are differentiated along a tholeiitic trend[1] during slow cooling. The variations in the chemistry of the principal elements are greater for the coarse grained rocks than for the basalts as shown on the alkali–iron–magnesium diagram[2]. Oceanic tholeiites are normally considered to have uniform chondrite normalized rare earth element (REE) patterns, characterized by La depletion, little or no Eu anomaly, and abundances ten to twenty-five times that of chondrites. The conditions leading to the largest variations of the principal elements in the intrusives should also lead to greater REE diversity. A progressive relative enrichment of Eu and depletion of La is found from granites and continental basalts through continental gabbros to oceanic tholeiites. These coarse grained rocks may increase this trend towards "deeper" depths, and thus we would predict a lower La/Sm ratio and larger positive Eu anomaly in the gabbros than in the oceanic basalt.

The samples were chosen to represent end members of tholeiitic and alkali differentiation (that is, those that are iron, alkali and magnesium-rich). The detailed petrology and inferred stratigraphic relationships of the samples are discussed in a series of papers by Bonatti *et al.*[2–4]. Table 1 shows location, abundances of the principal elements and petrographic data for the samples. The sampling of a small fragment, which may contain an anomalously high percentage of trace minerals or of the outer crust, may lead to REE abundances uncharacteristic of the whole rock. To minimize such influences, large samples (300 to 1,000 g) were cut from the interior, crushed, ground and then split for analysis.

The analytical procedure used was that of Schilling[5]. The REE were chemically isolated from 0.5 g of sample and the process monitored by ^{139}Ce and ^{88}Y tracers. After irradiation and chromatographic separation the fractions were gross beta counted in a gas flow proportional counter, and chemical yield and flux corrections applied. Accuracy and precision varied but in general were in the range of 4% to 10% (Table 2).

A triangular diagram permits simple recognition of differences in large numbers of relative REE distribution patterns. Triangular diagrams have been used to interpret the REE pattern relationships of samples from REE ore deposits of

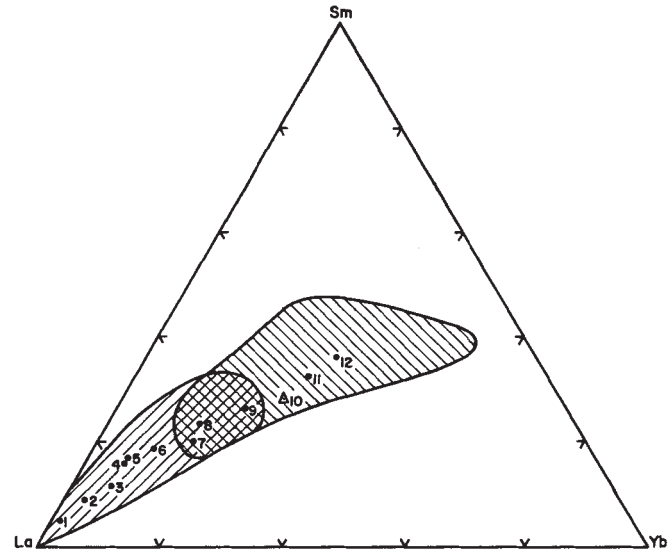


Fig. 1 Rare earth element relative distributions in igneous rocks. 1, Trachyte, Westerwold[15]; 2, trachyte, average of Gough and Ascension Is.[16]; 3, granite, 489 samples[15]; 4, continental basalt, 282 samples[15]; 5, intermediate rocks less than 60% silica[8]; 6, andesites, SW Pacific Islands[17]; 7, diabase and dolerites[18]; 8, gabbros, norites and gabbrodiorites[17]; 9, gabbros, Irontown, San Marcos, Skaeragaard, and 24 Finnish[15]; 10, chondrites, average of 26 including composite of 9 (ref. 15); 11, lunar basalt, average of types A, B, C and D, Apollo 11 (ref. 19); 12, oceanic basalt, average of 10 (ref. 15); shading indicates the approximate demarcation between oceanic floor and continental samples.

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